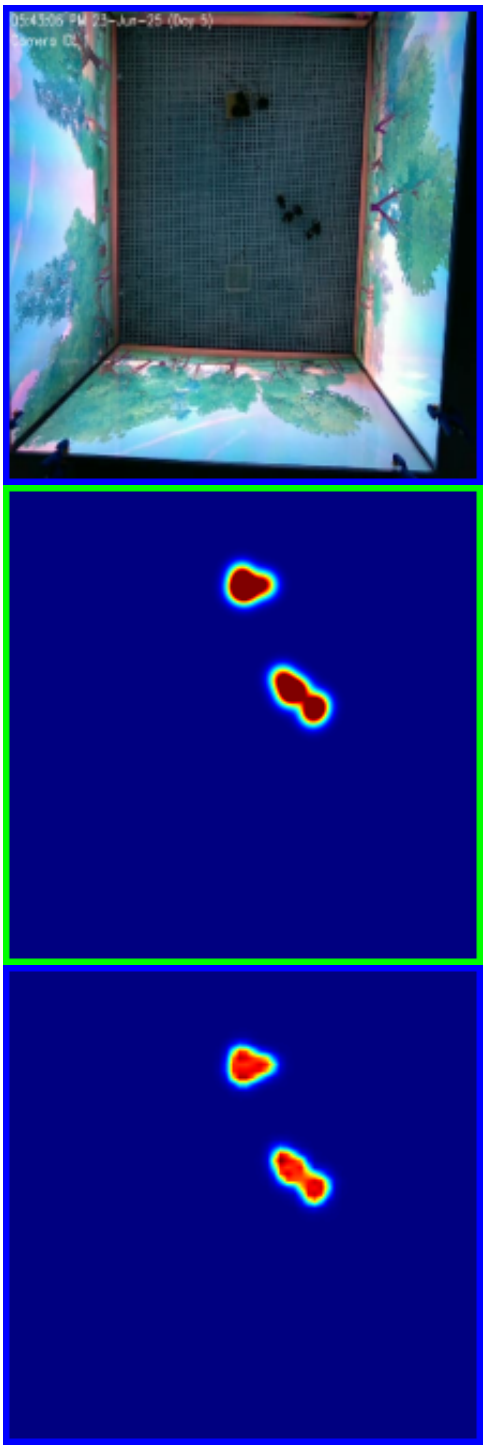
Background Mask

The animal tracker works through two main components: a background segmenter and a keypoint predictor. The background segmenter handles the first step by creating a mask that outlines the chicks and separates them from everything else in the frame. This mask is crucial because it narrows down where the keypoint predictor needs to look, making the whole system faster and more accurate.

The images on the left shows debug output from the background segmenter during training. These images help us see how well the model is learning. At the top is the current training frame, which is the raw image the model is working with. Below that, the green-framed image shows the ground truth mask. This is our reference point, showing what a perfect mask should look like for that particular frame.

The blue-framed image at the bottom displays what the model actually predicted. When you compare it to the ground truth above, you can see they look very similar. This close match tells us the model has successfully learned to tell the difference between chicks and background elements like the floor, walls, and equipment.

To measure how well the model is doing, we compare its predictions against the ground truth and calculate accuracy scores. These metrics let us track improvement over time. The model goes through this training cycle repeatedly, learning and adjusting with each pass. This particular background segmenter was trained over 120 epochs, meaning it saw the training data and refined its predictions 120 times before reaching its current performance level.

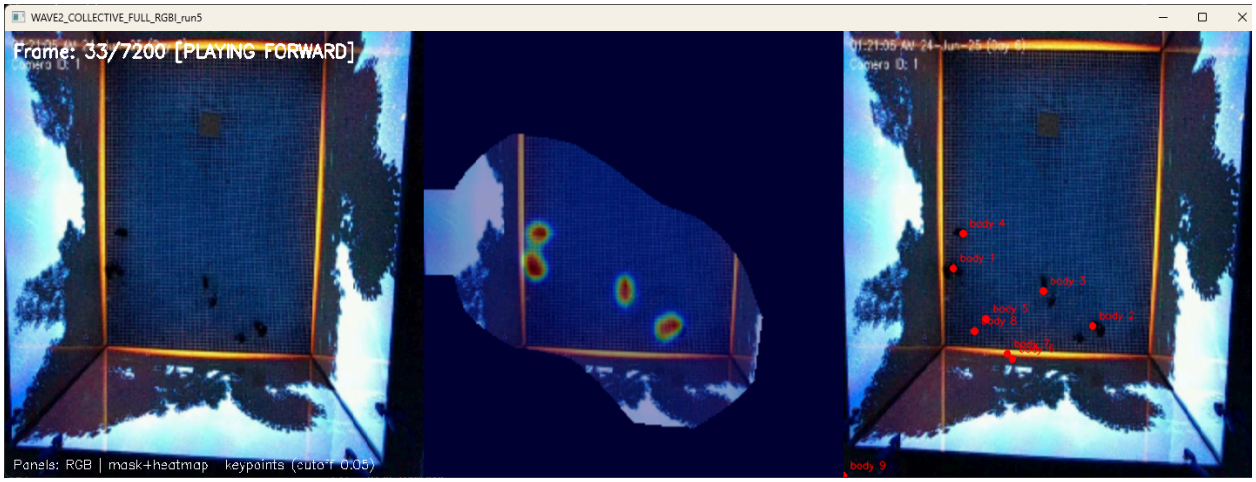


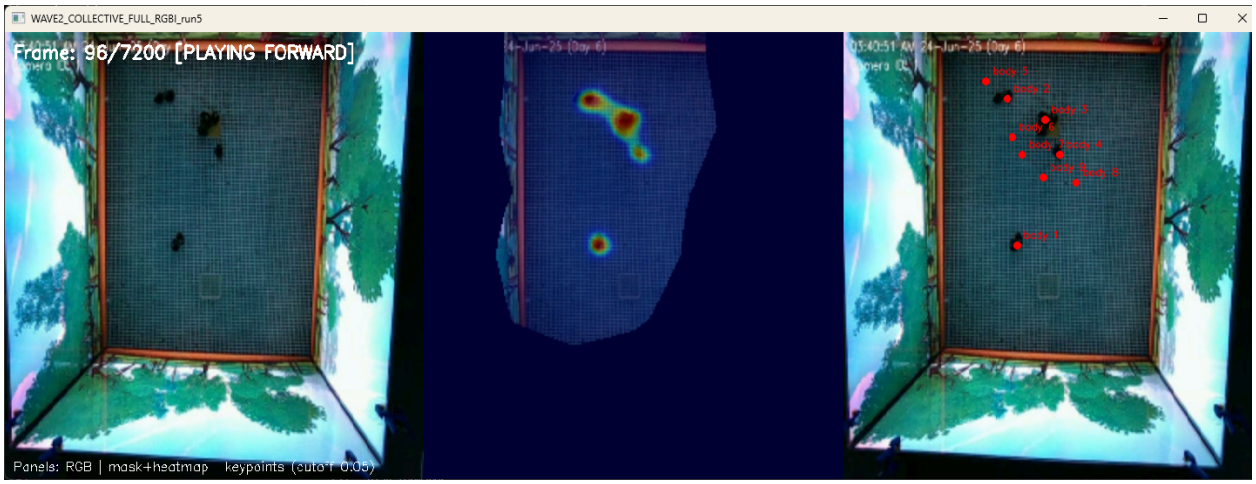
Keypoint Predictor

The keypoint predictor takes the masked image and scans through it to place a keypoint on each chick. Finding the right model for this step was harder than expected, as several factors make accurate predictions tricky. When chicks huddle together in groups, it becomes difficult to tell where one chick ends and another begins. The same problem happens when chicks position themselves close to the screen walls, where the overlap makes it tough to identify individual birds.

The images on the left follow the same layout as the background segmenter: the top shows the original training frame, the green-bordered image shows the ground truth, and the blue-framed image shows what the model predicted. Looking at the heatmap the model creates, you can see large peaks where the chicks are located. This part works well, and the model does a good job identifying that chicks are present in specific areas.

The real challenge comes next. Once the model identifies these hot zones, it needs to figure out how many chicks are actually in each one. Is that bright spot on the heatmap one chick or three chicks pressed together? This is where most models we tested ran into problems. While they could generate accurate heatmaps showing where activity was happening, they struggled to count the individual chicks within those areas.





Real Time Inference

This is the final result combining both models, processing video input to make predictions. What makes this exciting is the shift from recorded footage to real-time analysis. You can now point a camera at the chicks, and the model instantly generates the predictions shown in the middle and right images. The middle image shows how the two models work together. First, they create a background mask that separates the chicks from their surroundings. Then they generate a heatmap highlighting where the chicks are located within that mask. The right image demonstrates the next step: the model places keypoints–shown as red dots–marking each chick's position .The predictions aren't perfect yet, but they've come a long way since we started this project. The current accuracy level works well enough for practical use. Being able to track the chicks in real time opens up new possibilities that weren't available when we could only analyze pre-recorded footage.